Matrices (AI HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Matrix Multiplication and Inverses**

A transition matrix is $T = \left(\frac{4}{5}, \frac{3}{5}; \frac{1}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|B)$, the probability of moving from state B to state B.

Final Answer: $\frac{2}{5}$. Working through this additional-topics step:

Reading the matrix entry for $B \to B$ gives $\frac{2}{5}$.

2 **[Foundational]** • **Target: Determinants and Singularity**

A transition matrix is $T = \left(\frac{4}{5}, \frac{2}{5}; \frac{1}{5}, \frac{3}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.

Final Answer: $\frac{4}{5}$. Working through this additional-topics step:

Reading the matrix entry for $A \to A$ gives $\frac{4}{5}$.

3 **[Foundational]** • **Target: Eigenvalues and Eigenvectors**

A transition matrix is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{3}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

Final Answer: $\frac{3}{5}$. Working through this additional-topics step:

Reading the matrix entry for $A \to B$ gives $\frac{3}{5}$.

4 **[Foundational]** • **Target: Diagonalization & Matrix Powers**

A transition matrix is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{3}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.

Final Answer: $\frac{2}{5}$. Working through this additional-topics step:

Reading the matrix entry for $A \to A$ gives $\frac{2}{5}$.

5 **[Foundational]** • **Target: Matrix Multiplication and Inverses**

A transition matrix is $T = \left(\frac{4}{5}, \frac{4}{5}; \frac{1}{5}, \frac{1}{5}\right)$, rows/cols ordered A, B. Find $P(A|B)$, the probability of moving from state B to state A.

Final Answer: $\frac{4}{5}$. Working through this additional-topics step:

Reading the matrix entry for $B \to A$ gives $\frac{4}{5}$.

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6 [Foundational] • Target: Determinants and Singularity

A transition matrix is $T = \left(\frac{2}{5}, \frac{4}{5}; \frac{3}{5}, \frac{1}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.

Final Answer: $\frac{2}{5}$. Working through this additional-topics step:

Reading the matrix entry for $A \to A$ gives $\frac{2}{5}$.

7 [Foundational] • Target: Eigenvalues and Eigenvectors

A transition matrix is $T = \left(\frac{3}{5}, \frac{1}{5}; \frac{2}{5}, \frac{4}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

Final Answer: $\frac{2}{5}$. Working through this additional-topics step:

Reading the matrix entry for $A \to B$ gives $\frac{2}{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Diagonalization & Matrix Powers

A transition matrix is $T = \left(\frac{3}{5}, \frac{1}{5}; \frac{2}{5}, \frac{4}{5}\right)$, rows/cols ordered A, B. Find $P(B|B)$, the probability of moving from state B to state B.

Final Answer: $\frac{4}{5}$. Working through this additional-topics step:

Reading the matrix entry for $B \to B$ gives $\frac{4}{5}$.

9 [Intermediate] • Target: Matrix Multiplication and Inverses

A transition matrix is $T = \left(\frac{1}{5}, \frac{2}{5}; \frac{4}{5}, \frac{3}{5}\right)$, rows/cols ordered A, B. Find $P(A|B)$, the probability of moving from state B to state A.

Final Answer: $\frac{2}{5}$. Working through this additional-topics step:

Reading the matrix entry for $B \to A$ gives $\frac{2}{5}$.

10 [Intermediate] • Target: Determinants and Singularity

A transition matrix is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{1}{5}, \frac{4}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

Final Answer: $\frac{1}{5}$. Working through this additional-topics step:

Reading the matrix entry for $A \to B$ gives $\frac{1}{5}$.

11 [Intermediate] • Target: Eigenvalues and Eigenvectors

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{3}{10}, \frac{7}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

Final Answer: $\frac{3}{4}$. Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{3}{10}/(\frac{1}{10} + \frac{3}{10}) = \frac{3}{4}$

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12 [Intermediate] • Target: Diagonalization & Matrix Powers

A 2-state Markov chain has transition matrix $T = \left(\frac{7}{10}, \frac{3}{10}; \frac{1}{10}, \frac{9}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

Final Answer: $\frac{1}{4}$. Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{1}{10}/(\frac{3}{10} + \frac{1}{10}) = \frac{1}{4}$

13 [Intermediate] • Target: Matrix Multiplication and Inverses

A 2-state Markov chain has transition matrix $T = \left(\frac{7}{10}, \frac{3}{10}; \frac{3}{10}, \frac{7}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

Final Answer: $\frac{1}{2}$. Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{3}{10}/(\frac{3}{10} + \frac{3}{10}) = \frac{1}{2}$

14 [Intermediate] • Target: Determinants and Singularity

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{1}{10}, \frac{9}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

Final Answer: $\frac{1}{2}$. Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{1}{10}/(\frac{1}{10} + \frac{1}{10}) = \frac{1}{2}$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 5.02 Differential Equations

A 2-state Markov chain has transition matrix $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{1}{10}, \frac{9}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

Final Answer: $\frac{1}{5}$. Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{1}{10}/(\frac{2}{5} + \frac{1}{10}) = \frac{1}{5}$

16 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{3}{5}, \frac{2}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

Final Answer: $\frac{19}{25}$. Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{4}{5}, \frac{1}{5}; \ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{19}{25}$

17 [Challenging] • Target: Matrix Multiplication and Inverses • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{2}{5}, \frac{3}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

Final Answer: $\frac{13}{25}$. Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{3}{5}, \frac{2}{5}; \ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{13}{25}$

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18 [Challenging] • Target: Determinants and Singularity • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{2}{5}, \frac{3}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

Final Answer: $\frac{18}{25}$. Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{4}{5}, \frac{1}{5};\ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{18}{25}$

19 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 6.03 Markov Chains (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{4}{5}, \frac{1}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

Final Answer: $\frac{17}{25}$. Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{3}{5}, \frac{2}{5};\ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{17}{25}$

20 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 1.01 Proof by Mathematical Induction

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{4}{5}, \frac{1}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

Final Answer: $\frac{16}{25}$. Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{2}{5}, \frac{3}{5};\ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{16}{25}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Matrix Multiplication and Inverses

Compute the matrix product AB for $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$.

a. $[[4, 6], [10, 12]]$ ✓

b. $[[2, 0], [3, 12]]$

c. $[[4, 10], [6, 12]]$

d. $[[3, 2], [4, 7]]$

Row 1: $[1(2) + 2(1), 1(0) + 2(3)] = [4, 6]$. Row 2: $[3(2) + 4(1), 3(0) + 4(3)] = [10, 12]$.

22 [Foundational] • Target: Matrix Multiplication and Inverses

Find the inverse of the matrix $M = [[3, 5], [1, 2]]$.

a. $[[2, -5], [-1, 3]]$ ✓

b. $[[-2, 5], [1, -3]]$

c. $[[3, -1], [-5, 2]]$

d. $[[1/3, 1/5], [1, 1/2]]$

$\det(M) = 3(2) - 5(1) = 6 - 5 = 1$. The inverse is $\frac{1}{\det}[[d, -b], [-c, a]] = [[2, -5], [-1, 3]]$.

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23 [Foundational] • Target: Matrix Multiplication and Inverses

If A and B are invertible matrices of the same dimension, what is $(AB)^{-1}$?

a. $B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}$

c. BA^{-1}

d. AB^{-1}

The socks-and-shoes property of matrix inverses reverses order: $(AB)^{-1} = B^{-1}A^{-1}$.

24 [Foundational] • Target: Determinants and Singularity

For what value of k is the matrix $A = [[k, 6], [2, 3]]$ singular (non-invertible)?

a. $k = 4$ ✓

b. $k = 6$

c. $k = 0$

d. $k = 9$

A matrix is singular when its determinant is zero: $\det(A) = 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

25 [Foundational] • Target: Determinants and Singularity

If A is a 3×3 matrix with $\det(A) = 5$, what is $\det(2A)$?

a. 40 ✓

b. 10

c. 20

d. 80

For an $n \times n$ matrix, $\det(cA) = c^n \det(A)$. Here $n = 3$, so $\det(2A) = 2^3 \times 5 = 8 \times 5 = 40$.

26 [Foundational] • Target: Determinants and Singularity

If $\det(A) = 4$, what is $\det(A^{-1})$?

a. $\frac{1}{4}$ ✓

b. -4

c. 4

d. 16

$\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{4}$.

27 [Foundational] • Target: Eigenvalues and Eigenvectors

Find the eigenvalues of the matrix $A = [[4, 2], [1, 3]]$.

a. $\lambda = 5$ **and** $\lambda = 2$ ✓

b. $\lambda = 4$ and $\lambda = 3$

c. $\lambda = 6$ and $\lambda = 1$

d. $\lambda = 7$ and $\lambda = 0$

Characteristic equation: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$. Roots are $\lambda = 5, 2$.

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28 [Foundational] • Target: Eigenvalues and Eigenvectors

Find an eigenvector corresponding to $\lambda = 5$ for $A = [[4, 2], [1, 3]]$.

a. $[2, 1]^T$ ✓

b. $[1, 2]^T$

c. $[1, -1]^T$

d. $[2, -1]^T$

Solve $(A - 5I)\mathbf{v} = \mathbf{0} \Rightarrow [[-1, 2], [1, -2]][x, y]^T = [0, 0]^T \Rightarrow -x + 2y = 0 \Rightarrow x = 2y$. With $y = 1$, $\mathbf{v} = [2, 1]^T$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Eigenvalues and Eigenvectors

What is the sum of the eigenvalues of an $n \times n$ matrix A equal to?

a. **The trace of A (sum of diagonal entries)** ✓

b. The determinant of A

c. The rank of A

d. n

By algebraic properties of the characteristic polynomial, $\sum \lambda_i = \text{tr}(A)$.

30 [Intermediate] • Target: Diagonalization & Matrix Powers

If a matrix A is diagonalized as $A = PDP^{-1}$, how is the power A^n computed efficiently?

a. $A^n = PD^nP^{-1}$ ✓

b. $A^n = P^nD^n(P^{-1})^n$

c. $A^n = D^n$

d. $A^n = PDP^{-1}n$

$A^n = (PDP^{-1})(PDP^{-1}) \ldots = PD^nP^{-1}$. For a diagonal matrix D, D^n simply raises each diagonal entry to the nth power.

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31 [Intermediate] • Target: Diagonalization & Matrix Powers

If $D = [[2, 0], [0, 3]]$, what is D^4?

a. $[[16, 0], [0, 81]]$ ✓

b. $[[8, 0], [0, 12]]$

c. $[[24, 0], [0, 24]]$

d. $[[16, 81], [0, 0]]$

For diagonal matrices, powers are computed element-wise along the diagonal: $2^4 = 16$ and $3^4 = 81$, yielding $[[16, 0], [0, 81]]$.

32 [Intermediate] • Target: Diagonalization & Matrix Powers

When is an $n \times n$ matrix A guaranteed to be diagonalizable?

a. **When it has n linearly independent eigenvectors (or n distinct eigenvalues)** ✓

b. When its determinant is non-zero

c. Only if it is a symmetric identity matrix

d. When all its entries are positive

A matrix is diagonalizable if and only if there exists an eigenbasis consisting of n linearly independent eigenvectors.

33 [Intermediate] • Target: Matrix Multiplication and Inverses

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{1}{5}, \frac{4}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{2}{3}$ ✓

b. $\frac{2}{5}$

c. $\frac{9}{10}$

d. $\frac{11}{20}$

Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{1}{5}/(\frac{1}{10} + \frac{1}{5}) = \frac{2}{3}$

34 [Intermediate] • Target: Determinants and Singularity

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{2}{5}, \frac{3}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{3}{10}$

b. $\frac{1}{4}$

c. $\frac{1}{2}$

d. $\frac{4}{5}$ ✓

Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{2}{5}/(\frac{1}{10} + \frac{2}{5}) = \frac{4}{5}$

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35 **[Intermediate]** • **Target: Eigenvalues and Eigenvectors**

A 2-state Markov chain has transition matrix $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{1}{5}, \frac{4}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{1}{3}$ ✓

b. $\frac{4}{5}$

c. 1

d. $\frac{3}{10}$

Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{1}{5}/(\frac{2}{5} + \frac{1}{5}) = \frac{1}{3}$

36 **[Intermediate]** • **Target: Diagonalization & Matrix Powers**

A 2-state Markov chain has transition matrix $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{2}{5}, \frac{3}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{1}{2}$ ✓

b. $\frac{3}{10}$

c. 1

d. $\frac{2}{5}$

Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{2}{5}/(\frac{2}{5} + \frac{2}{5}) = \frac{1}{2}$

37 **[Intermediate]** • **Target: Matrix Multiplication and Inverses**

A 2-state Markov chain has transition matrix $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{2}{5}, \frac{3}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{3}{5}$

b. $\frac{2}{5}$

c. $\frac{2}{3}$ ✓

d. 0

Working through this additional-topics step: $\text{Steady state}: \ \pi_1 = \frac{2}{5}/(\frac{1}{5} + \frac{2}{5}) = \frac{2}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Determinants and Singularity • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{3}{5}, \frac{2}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

a. $\frac{9}{10}$

b. $\frac{3}{5}$ ✓

c. 1

d. $\frac{2}{5}$

Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{3}{5}, \frac{2}{5};\ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{3}{5}$

39 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 6.03 Markov Chains (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{2}{5}, \frac{3}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

a. $\frac{2}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. 0

Working through this additional-topics step: $\pi_1 = \pi_0 T = \frac{2}{5}, \frac{3}{5};\ \pi_2 = \pi_1 T \Rightarrow P(A) = \frac{2}{5}$

40 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 1.01 Proof by Mathematical Induction

Eigenvalues of $(3, 1; 1, 3)$:

a. **2 and 4** ✓

b. 1 and 3

c. 3 and 3

d. 2 and 3

Applying AI HL — Additional Topics principles, $(3-\lambda)^2-1=0 \rightarrow \lambda=2,4$. Therefore, the verified correct option is 2 and 4.

41 [Challenging] • Target: Matrix Multiplication and Inverses • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 1 to vertex 1 (the $(1, 1)$ entry of A^2).

a. 1

b. 0

c. **2** ✓

d. 4

Computing A^2 gives entry $(1, 1) = 2$.

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42 [Challenging] • Target: Determinants and Singularity • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 2 to vertex 2 (the $(2, 2)$ entry of A^2).

a. **2** ✓

b. 3

c. 1

d. 5

Computing A^2 gives entry $(2, 2) = 2$.

43 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 1.04 Partial Fractions

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 2 to vertex 3 (the $(2, 3)$ entry of A^2).

a. 3

b. **1** ✓

c. 4

d. 2

Computing A^2 gives entry $(2, 3) = 1$.

44 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 3 from vertex 2 to vertex 3 (the $(2, 3)$ entry of A^3).

a. 6

b. 2

c. 0

d. **3** ✓

Computing A^3 gives entry $(2, 3) = 3$.

45 [Challenging] • Target: Matrix Multiplication and Inverses • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 3 from vertex 1 to vertex 1 (the $(1, 1)$ entry of A^3).

a. **2** ✓

b. 3

c. 0

d. 4

Computing A^3 gives entry $(1, 1) = 2$.

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